HOMEOMORPHISM ONTO A CLOSED SUBSET PLUS SURJECTIVE STRUCTURE MAP IS A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5056-5072, following the statement at 5043-5054.

Editorial note. The proof is an explicit chain of five named engines, each cited, ending in a definition check.

Allowed standard prerequisites (editorial).

- Closed subset of quasi-compact is quasi-compact. Topology, Lemma topology-lemma-closed-in-quasi-compact (cross-chapter, cited at ~5058). Condition (1) makes f quasi-compact.
- Injective on points with surjective stalks implies monomorphism. lemma-injective-points-surjective-stalks (same chapter, cited at ~5060). Conditions (1)(2) give that f is a monomorphism.
- Monomorphisms are separated. lemma-monomorphism-separated (same chapter, cited at ~5062). f separated, enabling the pushforward engine.
- Pushforward along qcqs preserves quasi-coherence. lemma-push-forward-quasi-coherent (card L26386, cited at ~5063). f_*O_X is quasi-coherent.
- Two out of three for quasi-coherence. lemma-extension-quasi-coherent (card L26381, cited at ~5067). Kernel of O_Y -> f_*O_X is quasi-coherent.
- Definition of closed immersion of locally ringed spaces. definition-closed-immersion-locally-ringed-spaces (same chapter, cited at ~5071). Concluding identification via locally generated kernel.

Source reference. Original definition 01HK: sources/stacks-schemes/pinned-original-source.tex, source lines 561–561. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01IE: sources/stacks-schemes/pinned-original-source.tex, source lines 1675–1675. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01L6: sources/stacks-schemes/pinned-original-source.tex, source lines 4828–4828. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01L4: sources/stacks-schemes/pinned-original-source.tex, source lines 4780–4780. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01LC: sources/stacks-schemes/pinned-original-source.tex, source lines 4987–4987. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01BE: focused-seeds/geometry/S9-schemes-characterize-closed-immersions/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 005C: focused-seeds/geometry/S9-schemes-characterize-closed-immersions/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Suppose that*

(1) *f induces a homeomorphism of X with a closed subset of Y, and*
(2) *$f^\sharp : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is surjective.*

Then f is a closed immersion of schemes.

Proof. Assume (1) and (2). By (1) the morphism f is quasi-compact (see Topology, Lemma 005C). Conditions (1) and (2) imply conditions (1) and (2) of Lemma 01L6. Hence $f : X \to Y$ is a monomorphism. In particular, f is separated, see Lemma 01L4. Hence Lemma 01LC above applies and we conclude that $f_*\mathcal{O}_X$ is a quasi-coherent $\mathcal{O}_Y$-module. Therefore the kernel of $\mathcal{O}_Y \to f_*\mathcal{O}_X$ is quasi-coherent by Lemma 01IE. Since a quasi-coherent sheaf is locally generated by sections (see Modules, Definition 01BE) this implies that f is a closed immersion, see Definition 01HK. □